

Online Learning with Semiparametric Stochastic Approximation

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- **Online Learning**

- Rapid growth driven by real-world applications and the digital economy.
- Processes data incrementally without storing the full dataset—ideal for large-scale problems.
- Real-time decision, enables immediate responses to new information as it arrives.

- **Challenges in Estimation and Inference with Streaming Data**

- Existing work is largely restricted to parametric models.
- Many problems involve:
 - Low-dimensional parameters of interest.
 - High-dimensional/nonparametric nuisance component.
- Semiparametric approaches remain underexplored.
- Inference: all intermediate estimates must be retained for variance calculation..

(Stochastic) Gradient Descent

A stochastic approximation method (Robbins and Monro [1951]), such as Stochastic Gradient Descent (SGD), is a salable algorithm for parameter estimation.

Let each observation at time t be $U_t = (y_t, x_t)$, where y_t is the response, $x_t \in \mathbb{R}^{d_1}$ is a low-dimensional covariate vector.

- Traditional SGD goal: find $\theta^* = \arg \min \mathbb{E}[f(\theta; U)]$
- $f(\cdot)$: (unknown) function, may corresponds to a squared loss function
- Iterative updating rule:

$$\theta_t = \theta_{t-1} - \eta_t \underbrace{\nabla f(\theta_{t-1}; y_t, x_t)}_{:=G(\theta_{t-1}; U_t)}, \quad (1)$$

recursively updates the estimate upon the arrival of each data point x_t , for $t = 1, \dots, T$.

Especially relevant for online learning

- η_t : learning rate at time t
- ∇f : gradient of $f(\cdot)$

Iterative algorithm converges to θ^* with high probability.

Adaptive learning in macroeconomics, $\eta_t = 1/t$.

The Framework

Propose Semi-SGD as a one-pass algorithm: low space and time complexity, requiring only the current data and the previous estimate.

Case 1. Consider the following model:

$$F_{y_t|x_t,v_t;\theta_\tau}^{-1}(\tau) = x_t^\top \theta_{\tau,1} + \underbrace{P^{k_t}(v_t)^\top \theta_{\tau,2}}_{:=\lambda_\tau(v_t)} + r_{\tau,t} \quad (2)$$

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Propose Semi-SGD as a one-pass algorithm: low space and time complexity, requiring only the current data and the previous estimate.

Case 1. Consider the following model:

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- Approximation $\lambda_\tau(v_t)$ using a sieve basis expansion $P^{k_t}(v_t)$.
- Denote $\theta_\tau = (\theta_{\tau,1}, \theta_{\tau,2})$, and $P^{k_t}(w_t) = (x_t, P^{k_t}(v_t))$ is the Sieve to use at time t .
- k_t can be pre-specified as a function of T , the terminal value of t . Abbreviate k_t as κ .

As a result, we can write down a semi-parametric stochastic approximation process as:

$$\theta_t = \theta_{t-1} - \eta_t P^\kappa(w_t)(\tau - 1(P^\kappa(w_t)^\top \theta_{t-1} - y_t < 0)), \quad (3)$$

where we focus on $\eta_t = \eta_0 t^{-\alpha}$ as the learning rate, with $\eta_0 > 0$, and $\alpha \in (1/2, 1]$.

The Framework

Propose Semi-SGD as a one-pass algorithm: low space and time complexity, requiring only the current data and the previous estimate.

Case 2. Consider the following model:

$$y_t = x_t^\top \theta_1 + \underbrace{P^{k_t}(v_t)^\top \theta_2}_{:=\lambda(v_t)} + r_t + \varepsilon_t \quad (4)$$

- Approximation $\lambda(v_t)$ using a sieve basis expansion $P^{k_t}(v_t)$.
- Denote $\theta = (\theta_1, \theta_2)$, and $P^{k_t}(w_t) = (x_t, P^{k_t}(v_t))$ is the Sieve to use at time t .
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As a result, we can write down a semi-parametric stochastic approximation process as:

$$\theta_t = \theta_{t-1} - \eta_t P^\kappa(w_t) (P^\kappa(w_t)^\top \theta_{t-1} - y_t), \quad (5)$$

where we focus on $\eta_t = \eta_0 t^{-\alpha}$ as the learning rate, with $\eta_0 > 0$, and $\alpha \in (1/2, 1]$.

The Framework (Cont.)

Case 1. Given τ, κ , define $U_t = (y_t, w_t)$, and

$$G(\theta_{t-1}; U_t) = P^\kappa(w_t)^\top (\tau - 1(P_\kappa(w_t)^\top \theta_{t-1} - y_t < 0)). \quad (6)$$

Also, define $\theta^* := (\theta_{\tau,1}^*, \theta_{\tau,\kappa,2}^*)$ where

$$\theta_{\tau,\kappa,2}^* := \arg \min_{\theta_2} \|\lambda_\tau(v_t) - P^\kappa(v_t)^\top \theta_2\|_d \quad (7)$$

for some $\theta_2 \in \mathbb{R}^\kappa$.

The Framework (Cont.)

Case 2. Given κ , define $U_t = (y_t, w_t)$, and

$$G(\theta_{t-1}; U_t) = P^\kappa(w_t)^\top (P_\kappa(w_t)^\top \theta_{t-1} - y_t). \quad (8)$$

Also, define $\theta^* := (\theta_1^*, \theta_{\kappa,2}^*)$ where

$$\theta_{\kappa,2}^* := \arg \min_{\theta_2} \|\lambda(v_t) - P^\kappa(v_t)^\top \theta_2\|_d \quad (9)$$

for some $\theta_2 \in \mathbb{R}^\kappa$.

Illustrative Example of Motivation

Applicable to general control function approach in various models, e.g. Lee [2007]

$$y = x\beta_\tau + z_1^\top \gamma_\tau + u, \quad (10)$$

$$x = \mu(\alpha) + z^\top \pi(\alpha) + v, \quad (11)$$

and

$$Q_{u|x,z}(\tau) = \lambda_\tau(v). \quad (12)$$

The parameter of interest is the quantile parameter $\beta_\tau \in \mathbb{R}^p$ and $\gamma_\tau \in \mathbb{R}^{d_{z1}}$ for a specific value of quantile τ , with vector of exogenous explanatory variables $z \in \mathbb{R}^q$.

With a conditional independence condition that $u|v, z = u|v$, it can be shown that

$$F_{y|x,z;\theta}^{-1}(\tau) = x\beta_\tau + z_1' \gamma_\tau + \lambda_\tau(v) \quad (13)$$

where $\lambda_\tau(v)$ is a unknown non-parametric function of v . Here, $\theta_{\tau,1} = (\beta_\tau, \gamma_\tau)$

Relation to Literature (very selective)

- Traditional stochastic gradient descent algorithm
 - Robbins and Monro [1951], Kiefer and Wolfowitz [1952], Ruppert [1988], Polyak and Juditsky [1992].
- Online learning
 - Bottou et al. [1998], Mairal, Bach, Ponce, and Sapiro [2010], Hoffman, Bach, and Blei [2010].
 - Zhang and Simon [2022], nonparametric but no inference
- Recent work focusing on inferences for parametric models
 - Chen, Liu, and Zhang [2021], Li, Liu, Kyrillidis, and Caramanis [2018], Forneron [2022], Lee, Liao, Seo, and Shin [2022], Fang, Xu, and Yang [2018].

Algorithm 1: Semi-SGD for SQR as in (5)

Input : Function

Initialization: Set θ , κ , B and T

for $t = 1, \dots, T$ **and** $b = 1, \dots, B$ **do**

 for any positive integer κ , construct $P^\kappa(w_t) = [x_t, p_{1\kappa}(v_t), \dots, p_{\kappa\kappa}(v_t)]$ and update θ (and θ^b) via

$$\theta_t = \theta_{t-1} - \eta_t \cdot P^\kappa(\omega_t)(\tau - \mathbf{1}(P^\kappa(\omega_t)^\top \theta_{t-1} - y_t < 0)),$$

$$\theta_t^b = \theta_{t-1}^b - \eta_t \cdot W_{t,b} \cdot P^\kappa(\omega_t)(\tau - \mathbf{1}(P^\kappa(\omega_t)^\top \theta_{t-1} - y_t < 0)),$$

 where η_t and $W_{t,b}$ are the step sizes (learning rates) and bootstrap weights of the t -th update respectively.

end

Output : obtain $(1-\alpha)$ -confidence interval estimator of $\bar{\theta}_t$: $\bar{\theta}_t \pm z_{\alpha/2} \tilde{\sigma}_B$, where $\tilde{\sigma}_B$ obtained from the bootstrap procedure.

Algorithm 2: Semi-SGD Control Function Approach for endogenous QR as in (13), given τ

Input : Function
Initialization : Set θ , T_1 , κ , B, and T
Step 1 (the offline CF-QR) : for $t = 1, \dots, T_1$ do
 Observe $(y_{1:T_1}, x_{1:T_1}, z_{1,1:T_1}, z_{2,1:T_1})$
 Step 1a: Run QR of $x_{1:T_1}$ on $(1, z_{1,1:T_1}, z_{2,1:T_1})$ get π_{T_1} and $v_{1:T_1}$ from eq (11)

 Step 1b: Given $\hat{v}_{1:T_1}$ as estimates of $v_{1:T_1}$, consider a series regression with $w_{1:T_1} = (x_{1:T_1}, z_{1,1:T_1}, P^\kappa(\hat{v}_{1:T_1}))$ as covariates, where $P^\kappa(\hat{v}_{1:T_1})$ is a Sieve of $\hat{v}_{1:T_1}$;
 Run QR again of $y_{1:T_1}$ on $w_{1:T_1}$, and obtain estimates of $(\hat{\beta}_{\tau, T_1}, \hat{\gamma}_{\tau, T_1})$
end
Step 2 (the online semi-SGD part): for $t = T_1 + 1, \dots, T$ do
 Step 2a: Given quantile index α , update π and v

$$\pi_t = \pi_{t-1} - \eta_{1t} \cdot z_t^\top (\alpha - \mathbf{1}(z_t^\top \pi_{t-1} - x_t < 0));$$

$$v_t = x_t - z_t^\top \pi_t$$

 Step 2b: for any positive integer κ , construct $P^\kappa(w_t) = [x_t, z_{1t}, p_1(v_t), \dots, p_\kappa(v_t)]$; and update θ

$$\theta_t = \theta_{t-1} - \eta_{2t} \cdot P_\kappa(w_t)(\tau - \mathbf{1}(P_\kappa(w_t)^\top \theta_{t-1} - y_t < 0));$$

end
 η_{1t} and η_{2t} are the step sizes (learning rates) of the t -th update for Step 1 and Step 2 respectively.

- Allow for κ increase at each step
- For the Initial Step in Algorithm 2, we are using T_1 observations to get good initial estimates for $\lambda_\tau(v)$
- For asymptotic results, discuss two cases, $\alpha \in (1/2, 1)$, and $\alpha = 1$
- stochastic (sub)gradient descent

Assumptions

- 1 (a) Data $U_t = \{(y_t, x_t, v_t), t = 1, \dots, T\}$ are independently distributed. x_t, v_t has bounded and compact support $\mathcal{X} \times \mathcal{V}$; (b) $\lambda(v)$ is r -times continuously differentiable on $\mathcal{V}$.

- 2 Denote

$$A_\kappa := -\nabla \bar{G}(\theta_\kappa) = \mathbb{E}[P^\kappa(w_t)P^\kappa(w_t)^\top], \quad (14)$$

the Jacobian (Hadamard derivative) of the population gradient at θ_κ . Assume all eigenvalues of A_κ being positively bounded away from 0. Denote the lower bound is ψ .

- 3 For power series $\kappa = C_1 t^{v_1}$ for some constants C_1 satisfying $0 < C_1 < \infty$ and some v_1 satisfying $1/(2r) < v_1 < 1/8$, and for splines $\kappa = C_2 t^{v_2}$ for some constants C_2 satisfying $0 < C_2 < \infty$ and some v_2 satisfying $1/(2r) < v_2 < 1/5$.
- 4 (i) There exists a sequence $\zeta_0(\kappa)$ such that $\sup_{v \in \mathcal{V}} \|P^\kappa(v)\| \leq \zeta_0(\kappa)$, with $\zeta_0(\kappa)^2 \kappa / t^{\alpha/2} \rightarrow 0$.
(ii) $\|\lambda^*(\cdot) - P^\kappa(\cdot)^\top \theta_{\kappa,2}^*\| \leq C \kappa^{-\zeta}$ for some fixed constant $C > 0$ and $\zeta > 0$.

- Consider $\bar{G}(\theta) := \mathbb{E}[G(\theta; U_t)]$. It can be shown that

$$\bar{G}(\theta^*) = \mathbb{E}[P^\kappa(w_t)^\top (y_t - P^\kappa(w_t)^\top \theta^*)] \quad (15)$$

$$= \mathbb{E}[P^\kappa(w_t)^\top (x_t \theta_1^* + \lambda(v_t) - P^\kappa(w_t)^\top \theta_{\kappa,2}^*)] \quad (16)$$

$$= \mathbb{E}[P^\kappa(w_t)^\top (\lambda(v_t) - P^\kappa(v_t)^\top \theta_{\kappa,2}^*)] + O(\|\lambda(v_t) - P^\kappa(v_t)^\top \theta_{\kappa,2}^*\|^2), \quad (17)$$

under some regularity conditions, we can conclude that for each component of $\bar{G}(\cdot)$, we have that: $|\bar{G}_j(\theta^*)| \leq C' t^{-\nu'}$, $j = 1, 2, \dots, d_w$ for some fixed constant $C' > 0$.

Decomposition for fixed κ

$$\theta_t - \theta^* = \theta_{t-1} - \theta^* - \eta_t \bar{G}(\theta_{t-1}) - \eta_t (G(\theta_{t-1}; U_t) - \bar{G}(\theta_{t-1})) \quad (18)$$

$$= (I - \eta_t A_\kappa)(\theta_{t-1} - \theta^*) - \eta_t r(\theta_{t-1} - \theta^*) - \eta_t (G(\theta_{t-1}; U_t) - \bar{G}(\theta_{t-1})), \quad (19)$$

with $r(\theta - \theta^*) = \bar{G}(\theta) - \bar{G}(\theta^*) - A_\kappa(\theta - \theta^*)$ a high order residual.

Define $Q_{s,t} = \prod_{l=s}^{t-1} (I - \eta_l A_\kappa)$ which is a matrix discount factor. The updating condition can be written as:

$$\theta_t - \theta^* = \underbrace{Q_{0,t}(\theta_0 - \theta^*)}_{\Psi_1} - \underbrace{\sum_{s=1}^{t-1} \eta_s Q_{s,t} r(\theta_s - \theta^*)}_{\Psi_2} - \underbrace{\sum_{s=1}^{t-1} \eta_s Q_{s,t} (G(\theta_s, U_s) - \bar{G}(\theta_s))}_{\Psi_3}, \quad (20)$$

By construction, we have that $Q_{s,t} \leq \exp(-\frac{\eta_0 \psi}{1-\alpha}(t^{1-\alpha} - s^{1-\alpha}))$ for $\alpha \in (0, 1)$, and $Q_{s,t} \leq (\frac{s}{t})^{\eta_0 \psi}$ when $\alpha = 1$.

Lemma 1 [Risk Bound]

Define a metric, $\|\theta_t - \theta_\kappa\|_d^2 := \|\theta_{t,1} - \theta_1\|^2 + \mathbb{E}_{w_s}[|P(w_s)^\top(\theta_{t,2} - \theta_{\kappa,2})|^2]$.

When the learning rate is $\eta_t = \eta_0 t^{-\alpha}$, for t large enough, we have:

$$\mathbb{E}[\|\theta_t - \theta_\kappa\|_d^2] \leq \begin{cases} C_1 t^{-\alpha} \ln t, & \text{if } \alpha \in (\frac{1}{2}, 1), \\ C_2 t^{-1}, & \text{if } \alpha = 1 \text{ and } 2\psi\eta_0 > 1, \end{cases}$$

for some fixed positive constants $\eta_0, C_1, C_2 > 0$.

Asymptotics

Define $L_\kappa = \frac{1}{t} \sum_{s=1}^t P^\kappa(w_s) P^\kappa(w_s)^\top$. $A_\kappa^* := -\nabla \bar{G}(\theta_\kappa^*)$

When $\alpha < 1$,

$$\Sigma_\kappa := \eta_0 \int_0^\infty \exp(-u A_\kappa^*) L_\kappa \exp(-u A_\kappa^*)^\top du, \quad (21)$$

and when $\alpha = 1$,

$$\Sigma_\kappa := \eta_0 \int_0^\infty \exp(-u/\eta_0) \exp(-u A_\kappa^*) L_\kappa \exp(-u A_\kappa^*)^\top du. \quad (22)$$

The additional term $\exp(u/\eta_0)$ accounts for the linearly decaying step size.

Consider any C^1 functional $g(\theta_1^*, \lambda(\cdot))$ with bounded derivative that is approximated by $g(\theta_{1,t}, P^\kappa(\cdot)^\top \theta_{2,t})$. Denote

$\Omega_\kappa := \left(\frac{\partial b}{\partial \theta_{1,t}} \right)_{\frac{\partial b}{\partial \lambda} P^\kappa(\cdot)}$. That said, g is Hadamard differentiable with respect to λ , e.g.,

$$g(\theta_{1,t}, P^\kappa(v)^\top \theta_{2,t}) = a^\top \theta_{1,t} + \int_v P^\kappa(v)^\top \theta_{2,t} \mu(v) \quad (23)$$

for some probability measure $\mu(v)$.

Theorem 1. If all the assumptions above hold and: $\kappa^{-\zeta} t^{\alpha/2} \rightarrow 0$, $\zeta_0^2(\kappa) \kappa / t^{\alpha/2} \rightarrow 0$.

$$\sqrt{\eta_0 t^\alpha} (\Omega_\kappa^\top \Sigma_\kappa \Omega_\kappa)^{-\frac{1}{2}} (g(\theta_{1,t}, P_\kappa(\cdot)^\top \theta_{2,t}) - g(\theta_1^*, \lambda(\cdot))) \rightsquigarrow N(0, 1). \quad (24)$$

- DGP1: $Y = \mu + X_1\beta + X_2\gamma_1 + X_3 + X_4^2 + X_5^3 + X_6^4 + X_7^5 + U * (X_1\beta + X_2\gamma_1)$
- DGP2 (cf. Lee [2007]):

$$Y_i = X_i\beta + Z_{1i}\gamma + U_i, \quad U_i = V_i + \phi(V_i) + 0.5[\tilde{U}_i - F_{\tilde{U}}^{-1}(\tau)],$$
$$X_i = \mu + Z_{1i}\pi_1 + Z_{2i}\pi_2 + V_i, \quad V_i = \exp(Z_{2i}/2)\tilde{V}_i, \quad i = 1, \dots, n$$

where Z_{1i} , Z_{2i} , $\tilde{V}_i$ and $\tilde{U}_i$ are independently drawn from the standard normal distribution, $\phi(v) = 4 \exp[-(v-1)^2]$, and $F_{\tilde{U}}$ is the CDF of $\tilde{U}$. The function $\phi(v)$ has a bell-shaped hump around one and represents a nonlinear component of $\lambda_\tau(v) = v + \phi(v)$. We set the parameter values $(\beta, \gamma, \mu, \pi_1, \pi_2) = (1, 1, 1, 3, 1)$. In all experiments $\tau = 0.9$ and $\alpha = 0.5$.

Simulation Results

Figure 1: The simulation paths for SQR1 for $n = \{6000, 9000, 12,000\}$, $k = 3$, and coefficients $\{1, 0.5, 0.2\}$

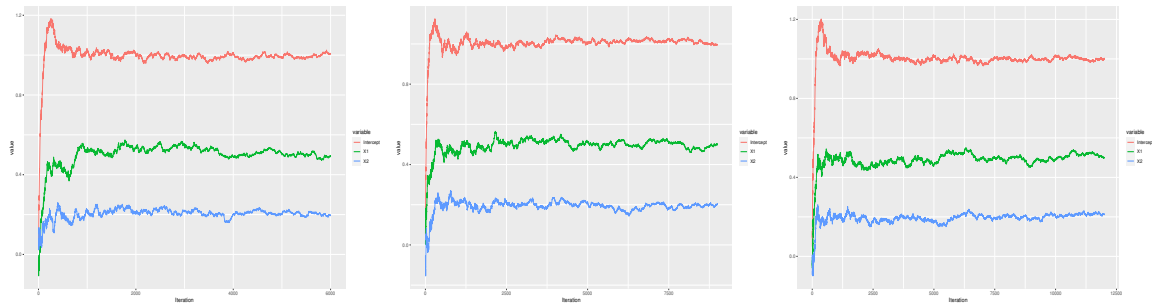


Table 1: Coverage Probabilities of 95% Confidence Intervals for Semiparametric QR

	(N, k, τ)		
	$(12000, 3, 0.5)$		
Bias	-0.003	0.0004	-0.0005
SE	0.004	0.008	0.005
CP	0.98	0.95	0.97
	$(12000, 4, 0.5)$		
Bias	-0.0005	0.0001	0
SE	0.004	0.008	0.005
CP	0.99	0.96	0.97
	$(12000, 5, 0.5)$		
Bias	-0.015	0.003	0.001
SE	0.01	0.01	0.01
CP	0.94	0.96	0.97
Based on 500 simulations, with $(\mu, \beta_1, \gamma_1) = (1, 0.5, 0.2)$			

Figure 2: The simulation paths for SQR1, $n = 12,000$, $k \in \{3, 4, 5\}$, and coefficients $\{1, 0.5, 0.2\}$

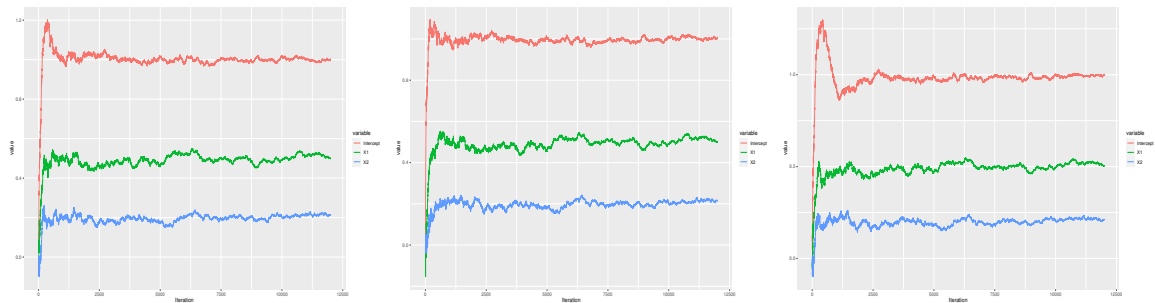
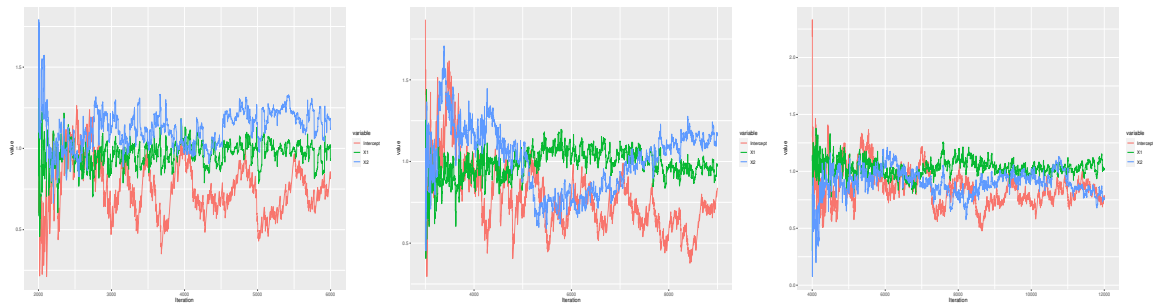


Figure 3: The simulation paths for SQR2, $n = \{6000, 9000, 12,000\}$, $k = 7$, and coefficients $\{1, 1, 1\}$



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